

Next Generation Ultra-High Performance Synergistic Imaging



PET

SPECT

Optical

CT

Why seeing more matters in molecular imaging

Multimodal synergy imaging: lifting *in vivo* imaging to the next level

MILabs is a leading technology innovator in preclinical imaging with PET, SPECT, Optical, and X-ray CT *in vivo* imaging modalities.

These systems have received many international awards from the molecular imaging community, and they support the research of scores of satisfied users worldwide, as evidenced by hundreds of peer-reviewed scientific publications. However, as we strive to meet your most demanding application needs, also these best-in-class imaging modalities must keep evolving.

Therefore, and truthful to our mission of “making molecular imaging clear”, MILabs has now succeeded at exploiting the multimodal synergy of its 4x4D imaging modalities by integrating them on a single scalable multimodality platform.

Now you have not only PET/CT or SPECT/CT imaging, but also simultaneous multi-isotope, PET, PET/PET, PET/SPECT and PET/Radiotherapy/CT imaging. Furthermore, in addition to 2D Optical imaging you can have CT-guided 3D Optical/CT tomography. All these capabilities are unique to MILabs' systems.

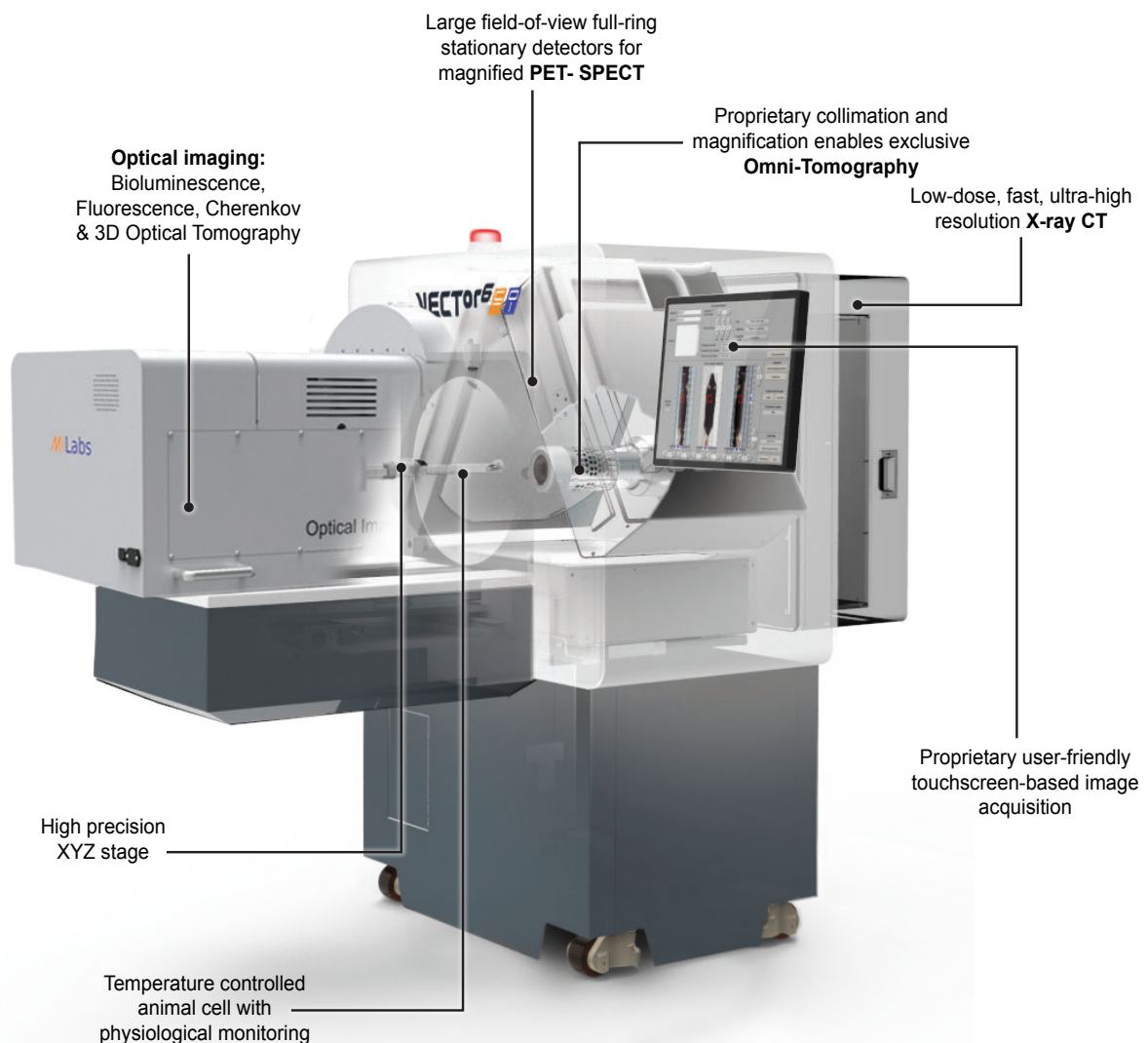
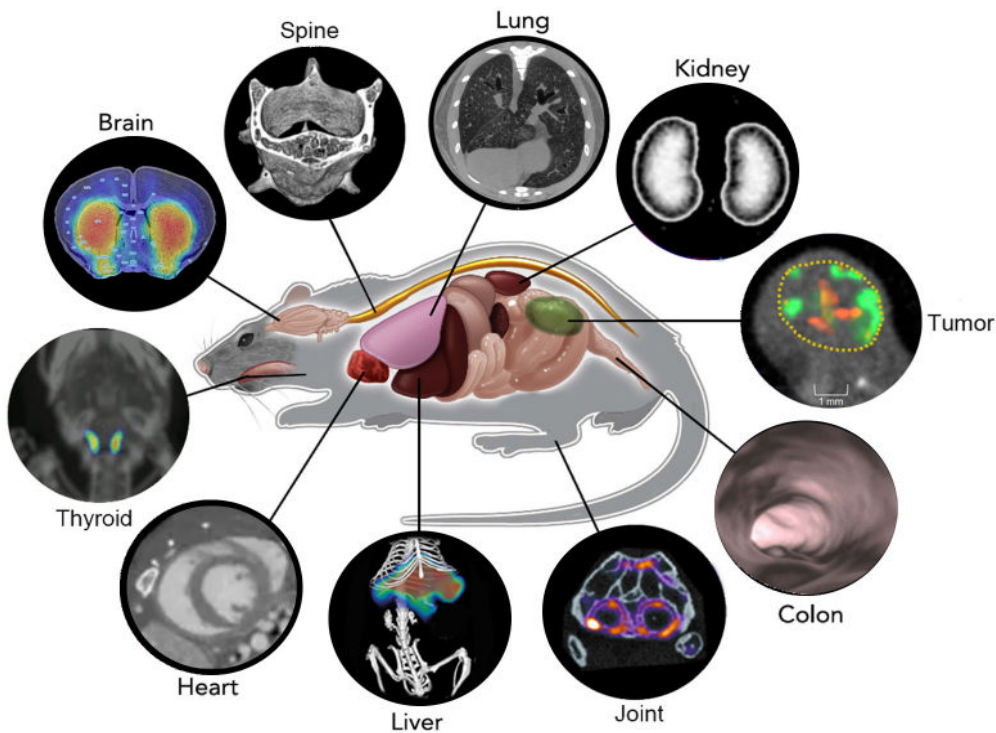


Table of contents

More Significant than the Sum of its Parts	2	PET & SPECT: VECTor [®] CT	12
Why seeing more matters in molecular imaging	4	U-OI: Optical Imaging & 3D Optical Tomography	14
Modular upgradable systems	5	• A scalable & highly flexible optical imaging platform	15
U-SPECT [®] CT	6	• Optical tomography	16
• Ultra-high resolution SPECT	7	U-CT: stand-alone or integrated X-ray CT	18
• EXIRAD-3D: Quantitative 3D autoradiography	8	• X-Ray CT beyond the application capabilities of other preclinical CT systems	20
• Clinical G-SPECT for large animals	9	Animal handling	22
U-PET [®] CT	10	Fully integrated software platform	23
• Why U-PET?	11	Awards and partnerships	24

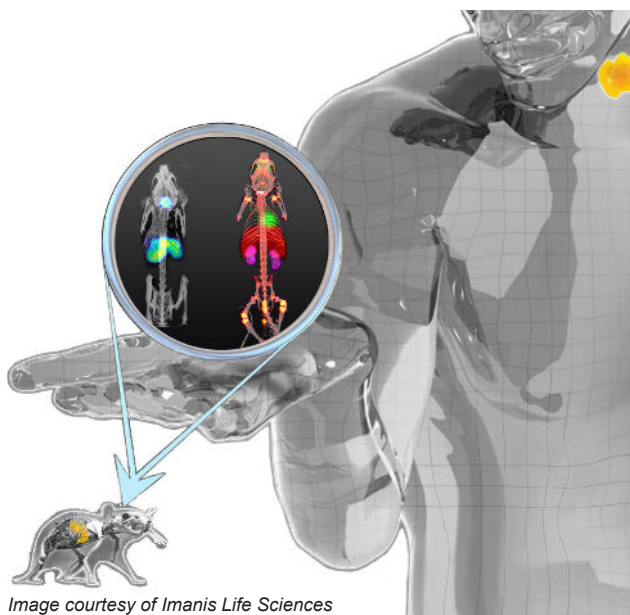
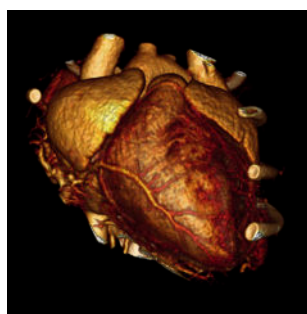
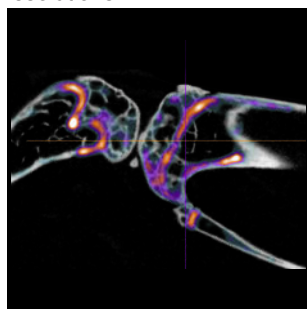


Value from state-of-the-art 4x4D multimodal synergy imaging

Through continuous innovation, we have now achieved a higher-order of multimodal imaging, called 4x4D multimodal synergy imaging. Four modalities, each with four-dimensional tomographic imaging capabilities, will enable you to improve diagnostics and therapy development through analysis of complementary, data-rich, fused images, many co-registered in both space and time. What's more, each modality by itself will give you data beyond the capabilities of any other standalone system.

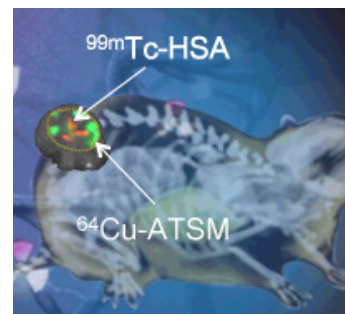
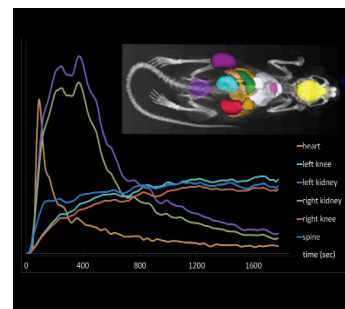
Breakthrough translational image resolution:

MILabs' systems have the ability to image and quantitate smaller structures better than any other nuclear imaging system. Imaging with SPECT down to 0.12 mm and PET down to 0.55 mm resolution give you the power to image mice with unprecedented visual acuity. Moreover, it is the only PET that can image ^{124}I and ^{82}Rb at sub-mm resolutions.



Higher sensitivity for 4D imaging applications:

The high sensitivity of MILabs' four imaging modalities enables you to obtain multiple registered 4D live streaming image videos, showing e.g., fast beating mouse hearts, blood flow in organs and vessels, the uptake and wash-out of molecular tracers, the biodistribution of new disease markers, and more.



Not just CT but Diagnostic CT:

MILabs' integrated adaptive diagnostic CT, extends the role of CT beyond co-registration of hybrid images. With the finest resolution, the lowest dose, and fastest acquisitions in the industry, MILabs X-ray CT adds many new diagnostic capabilities to your molecular research and includes dual-gated, multi-energy, and dynamic contrast-enhanced CT.

Multiple modalities working in unison:

With MILabs' multimodal synergy, you gain more by imaging less. Unique capabilities such as simultaneous multi-isotope PET and PET/SPECT acquisitions, high resolution theranostic/CT of α and β radiotherapies, and 3D Optical Tomography enable studying different functions at the same time, under identical physiological and physical conditions.



Example configurations:

VECTOr⁶CT: PET/SPECT/CT

- Integrated high performance PET/SPECT/CT
- Best PET resolution: ¹⁸F at 0.55 mm; ¹²⁴I at 0.75 mm
- Best SPECT resolution: 0.12 mm
- Dynamic PET & SPECT:
< 1 s organ and < 8 s total body frame rate
- Simultaneous multi-isotope PET/PET and PET-SPECT
- Radio-theranostics up to 1 MeV

U-CT: high-performance X-Ray CT

- Ultra-high resolution stand-alone or integrated CT
- Unprecedented scanning speed: 5 s total body imaging at low dose down to < 2 mGy
- Cardiac- and respiratory-gated 4D imaging
- Dynamic contrast-enhanced CT
- True diagnostic *in vivo* CT imaging

U-SPECT⁶CT

- Integrated high performance SPECT/CT
- Best SPECT resolution: 0.12 mm
- Dynamic SPECT:
< 1 s organ and < 8 s total body frame rate
- Ultra-low dose and quantitative SPECT
- Easily upgradeable with full PET capabilities
- High-energy SPECT

U-OI/CT

- Exclusive integrated Optical Imaging and high performance CT
- Bioluminescence, fluorescence, Cherenkov imaging
- 3D Optical tomography
- Removable Duet Optical Module for OI and OI/CT
- Docking station for powerful stand-alone OI imaging

Modular and field-upgradable

MILabs' systems have a modular set up where PET, SPECT, Optical or CT functionality can be delivered separately or in any combination. After initial installation, additional functionalities can be added to the platform.

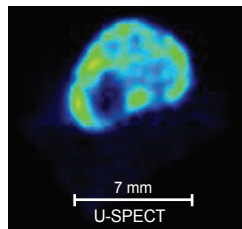
Scientifically validated ultra-high-resolution & fast dynamic SPECT imaging



Unmatched SPECT performance

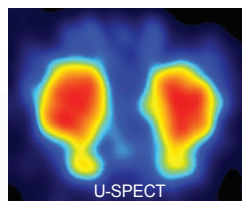
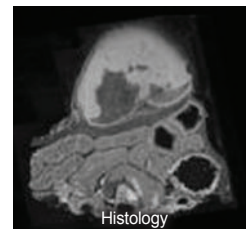
- Spatial resolution: proven best *in vivo* SPECT resolution in the market: < 0.25 mm [1]. Enables quantifying dynamic processes in tiny sub-compartments of mouse organs and tumors *in vivo*.
- 0.12 mm resolution 3D Autorad option (see p. 8).
- Temporal resolution: proven best SPECT total body and organ time resolution (< 8 s and < 1 s respectively). Perfect for assessing fast pharmacokinetics.
- Sensitivity: proven best sensitivity allows for < 0.10 MBq (~3 μ Ci) total body SPECT and fast screening of tracers. Up to 30,000 cps/MBq sensitivity.
- Reliability: patented robust stationary SPECT detector design ensures unprecedented reliability, virtually no down time and very long system life.
- Energy range: 20 keV - 1# MeV for single and multitracer SPECT.
- Quantitation: unrivaled accuracy due to advanced correction methods for scatter, attenuation, and partial volume effects.

U-SPECT

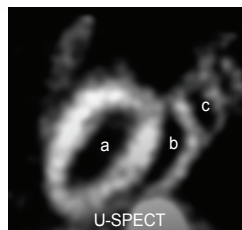
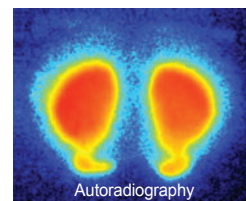


U-SPECT accurately shows heterogeneous antibody distribution within tumors [2].

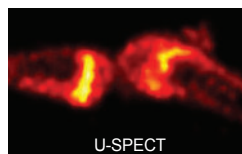
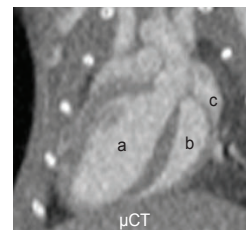
Reference



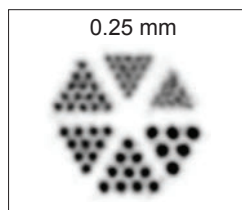
¹²³I-FP-CIT DAT SPECT of mouse brain vs. autoradiography [3].



Cardiac perfusion SPECT vs. μ CT shows a) left ventricle, b) right ventricle, and c) right atrium [4].



Quarter-mm resolution *in vivo* SPECT and high-resolution μ CT [1].

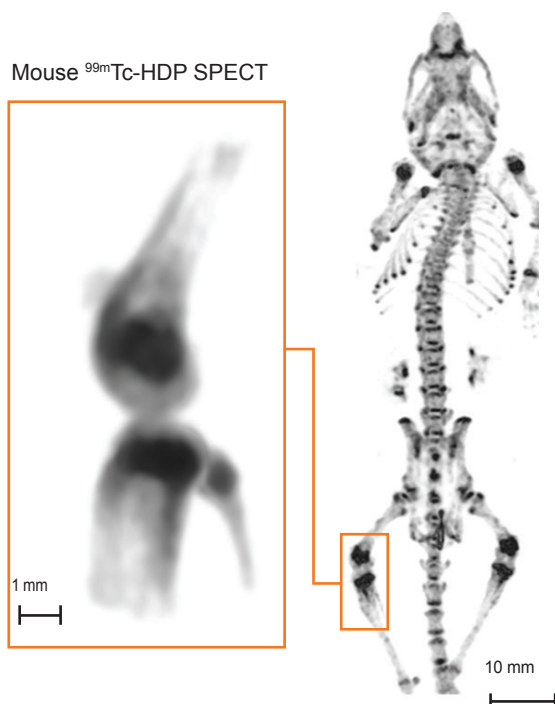


Validation of image resolution of *in vivo* U-SPECT [1].



1 Quarter-mm resolution molecular mouse imaging with U-SPECT[®], O. Ivashchenko et al., Mol. Imaging. 2014
 2 3D histological validation of high-resolution SPECT of antibody distributions within xenografts. W. Branderhorst, et al., J. Nucl. Med. 2014
 3 Movies of dopamine transporter occupancy with ultra-high resolution focusing pinhole SPECT. B. Vastenhouw, et al., Mol. Psychiatry 2007
 4 Comparison of 4D-MicroSPECT and MicroCT for Murine Cardiac Function, N.T. Befera, et al, Mol. Imaging Biol. 2013
 # With VECTor upgrade

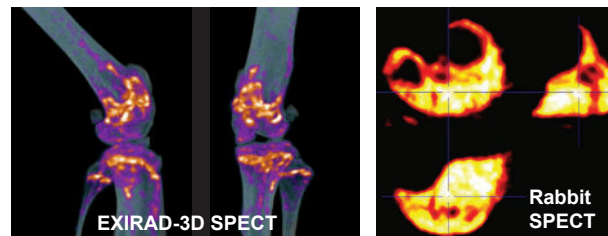
Ultra-high resolution SPECT



Better than a quarter-mm resolution *in vivo* SPECT imaging [1].

Small specimen to larger animals

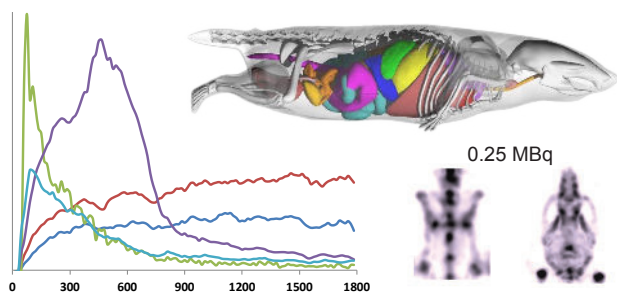
U-SPECT[®] offers the highest commercially available resolution for small specimens up to medium-sized animals, showing mouse bone scans down to 0.12 mm and rabbit liver scans at 2 mm resolutions [7].



Adaptive SPECT with proprietary multi-pinhole collimation

Superior system performance for a wide range of applications is achieved through MILabs' patented multi-pinhole collimators. A wide range of scientifically validated collimators is available to enable ultra-high resolution (0.12 mm) or ultra-sensitive (> 30,000 cps/MBq) imaging, providing incredibly detailed, low dose or ultra-fast imaging.

Ultra-fast and ultra-low dose SPECT



Quarter-minute SPECT imaging combined with quarter MBq (< 7 μCi dose) enables fast 4D pharmacokinetic analysis [5] and whole-body quick and easy *in vivo* biodistribution analysis [6].



Different sizes of collimators are available: 3D Autorad, mouse, mouse/rat and medium size animal (MSA) collimators.

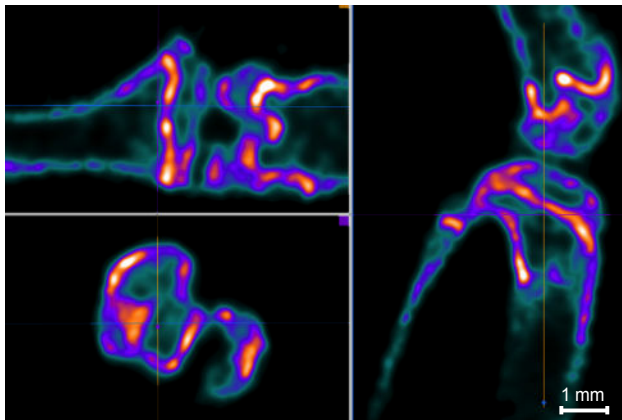
5 Whole-body sub-half-minute 4D pharmacokinetic SPECT in mice, R. Ramakers et al., J. Nucl. Med. 2013

6 Ultra-High-Sensitivity Submillimeter Mouse SPECT, O. Ivashchenko et al., J. Nucl. Med. 2015

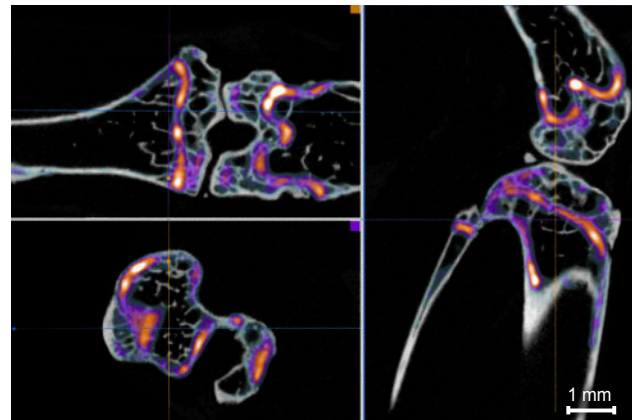
7 High-resolution medium-sized animal multi-pinhole SPECT with stationary detectors, F. van der Have et al., EANM 2014, Gothenburg, Sweden

By scanning cryo-cooled tissue samples, such as complete organs, with a dedicated super-resolution SPECT option, researchers can finally eliminate many tedious, time-consuming, and error-prone steps from quantitative 3D autoradiography.

Thanks to the excellent energy resolving capabilities of the detectors, one can scan multiple isotopes at the same time to image 3D distributions of different molecules simultaneously, with 0.12 mm resolution.



EXIRAD-3D SPECT

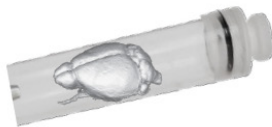


EXIRAD-3D SPECT/CT

Mouse knee imaged with super-resolution ^{99m}Tc -MDP SPECT (left) and ^{99m}Tc -MDP SPECT/CT (right) with high definition X-ray CT. These extreme resolution SPECT and CT capabilities enable studying the spatial distribution of a selected function marker and relate this one-to-one to anatomical microstructures imaged in 3D.



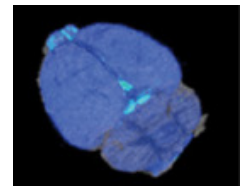
Fresh organ sample



EXIRAD-3D sample holder with cryo-cooling



EXIRAD-3D Automated 3D Autoradiography



Reconstructed quantitative 3D Autorad image

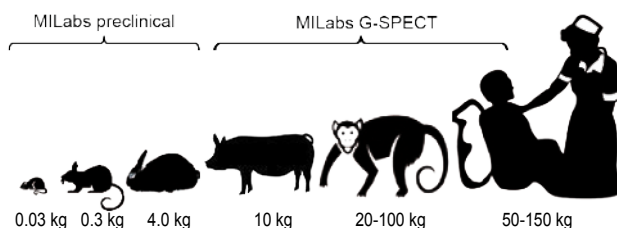
EXIRAD-3D multi-isotope autoradiography is fast, saves tremendous amounts of labor and reduces costs. Best of all, results are truly quantitative. On your image analysis workstation, undistorted quantitative EXIRAD-3D images can be re-sliced in any direction over and over, and can be perfectly overlaid with automatically registered

ultra-high resolution CT images obtained in the same multi-modal acquisition or with other 3D images of the specimen. With EXIRAD-3D, MILabs' U-SPECT[®]CT and VECTOR[®]CT systems offer a unique single-system solution to translate research results from *ex vivo* to *in vivo* analysis and vice-versa.

Gyro-free SPECT; from mouse to man



MILabs offers gyro-free SPECT for any size animal

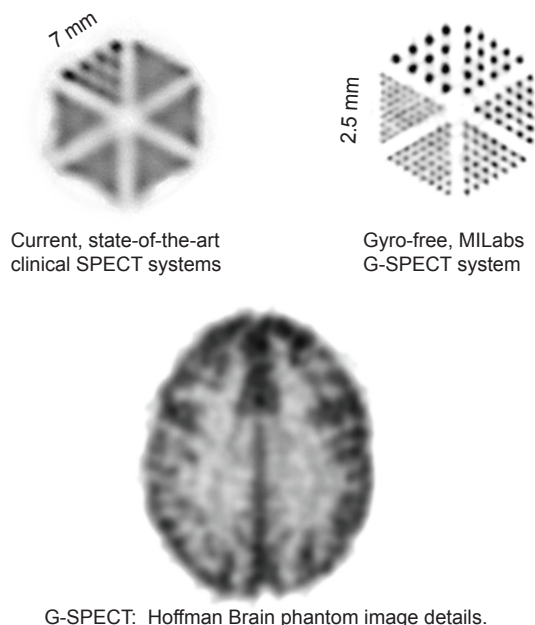


Unmatched SPECT performance

- Stunning image quality: with a ten-fold volumetric resolution improvement over current state-of-the-art large animal and clinical SPECT [8], G-SPECT outperforms dedicated PET systems for organ-specific imaging.
- Adaptive SPECT: user-exchangeable collimators allow for adaptation of resolution to any animal size to facilitate translation from mouse to large animal to humans without any information loss.
- Slash costs: thanks to the breakthrough full-ring, gyro-free, detector technology, G-SPECT can cut down tracer dose with a factor of ten without affecting image quality.
- Extra insight with dynamic SPECT: the high sensitivity of G-SPECT and the absence of detector rotation, enables a truly dynamic determination of tracer uptake.
- Reliable and stable: no wear-and-tear from rotating detectors and no instability or calibration problems caused by positioning inaccuracies.
- List-mode acquisitions: eliminate rescans caused by movement by retrospectively deleting the list data portion subject to motion

The award-winning large animal G-SPECT's design is based on the same proven gyro-free technology used in U-SPECT[®]. The result is a robust, fast, and cost-effective solution with unprecedented image quality for scanning large animals and humans.

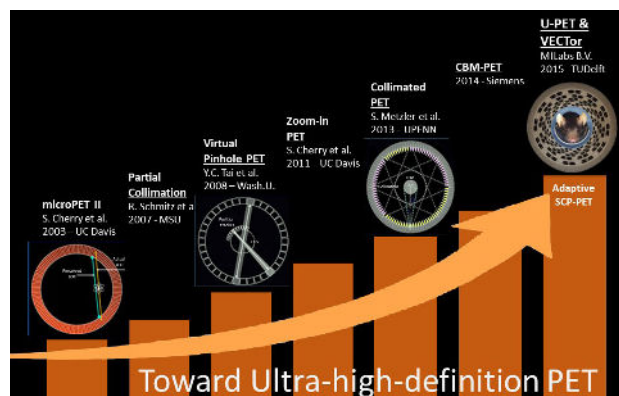
Clearer images at reduced doses



8 G-SPECT-I: a full ring high sensitivity and ultra-fast clinical molecular imaging system with < 3 mm resolution. Beekman F.J. et al. Eur. J. Nucl. Med. Mol. Imaging. 2015;42 Suppl 1:S209

* Clinical availability depends on final local registrations, including FDA and CE clearance.

Ultra-High-Definition PET ("U-PET"): best resolution and small lesion image contrast



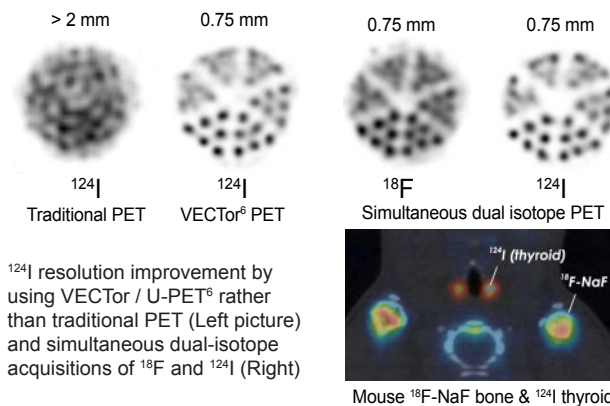
The evolution of *in vivo* preclinical PET is analogous to the microscopy evolution: "from wide-field microscopy to scanning confocal microscopy".

Unmatched PET performance

- Resolution: unrivaled uniform spatial resolution down to 0.55 mm for ^{18}F and still < 0.75 mm for isotopes with long positron range like ^{124}I and ^{82}Rb [9, 10].
- Best small lesion contrast and superior spatial resolution enables to use the U-PET system for a wide range of preclinical imaging applications [11].
- Dynamic PET: excellent organ & tissue-specific (< 1 s) and whole-body time resolution (< 8 s). Low-dose whole-body imaging (e.g. [12, 13, 14, 24])
- With VECTOR upgrade: unique sub-mm imaging for high energy radiotherapy isotopes: e.g. < 0.55 mm for ^{131}I [15], < 0.7 mm for ^{213}Bi [16] and < 0.85 mm for $^{211}\text{Rn}/^{211}\text{At}/^{209}\text{At}$ and ^{225}Ac [17].
- Extended application range: sub-mm resolution with improved S/N characteristics, enabling accurate quantitation at the organ substructure level. Performance will not be affected by co-injected SPECT tracers, enabling applications beyond the capabilities of conventional coincidence-based PET systems [18, 19].

Exclusive: Simultaneous dual-isotope PET and elimination of positron range blurring

- U-PET's superior performance for mouse imaging is further underscored by the ability to image isotopes, such as ^{124}I and ^{82}Rb without positron range blurring at 0.75 mm resolution. Moreover, you can image two co-injected PET tracers simultaneously such as ^{124}I /FDG for immuno-PET/ tumor-uptake and $^{82}\text{Rb}/^{18}\text{F}$ for myocardial perfusion/ viability studies.



Mouse ^{18}F -NaF bone & ^{124}I thyroid

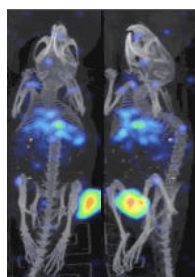
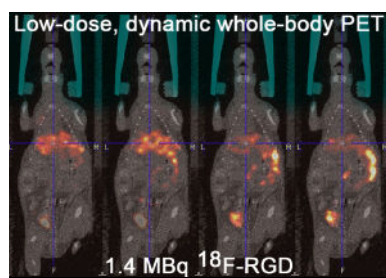
- 9 VECTOR-6: Broadband photon tomography takes the versatility of multi-modality preclinical imaging beyond the limits of PET-SPECT-CT and Optical Imaging. F.J. Beekman et al., WMIC 2018, WMIS Innovation of the Year, Seattle, WA, USA.
- 10 Zero Positron Range and Multi-Isotope Positron Emission Tomography. F.J. Beekman et al., WMIC 2018, Seattle, WA, USA.
- 11 Performance Assessment of a Preclinical PET Scanner with Pinhole Collimation by Comparison to a Coincidence-Based Small-Animal PET Scanner, M.D. Walker, et al., J. Nucl. Med. 2014
- 12 PET imaging of PARP expression using [^{18}F]olaparib. T. Wilson et al., J Nucl Med., 2018.
- 13 ^{18}F -Trifluoromethylation of Unmodified Peptides with 5- ^{18}F - (Trifluoromethyl)dibenzothienophenium Trifluoromethanesulfonate, S. Verhoog et al., J. Am. Chem. Soc. 2018
- 14 ^{18}F -FDG as an inflammation biomarker for imaging dengue virus infection and treatment response, A. Chacko et al., JCI Insight 2017
- 15 High-resolution Clustered-Pinhole ^{131}I SPECT Imaging in Mice, F. van der Have et al., Nucl. Med. Biol. 2016
- 16 Utilizing High-Energy γ -Photons for High-Resolution ^{213}Bi SPECT in Mice, J. de Swart et al., J. Nucl. Med. 2016
- 17 Multi-isotope SPECT imaging of the ^{225}Ac decay chain: feasibility studies, A. Robertson et al. Phys. Med. Biol., 2017
- 18 VECTOR: A Pre-clinical Imaging System for Simultaneous Sub-mm SPECT-PET, M.C. Goorden, et al., J. Nucl. Med., 2013
- 19 Performance Characteristics of a Novel Clustered Multi-pinhole Technology for Simultaneous High-resolution SPECT/PET, K. Miwa et al., Ann. of Nucl. Med. 2015

Why U-PET?

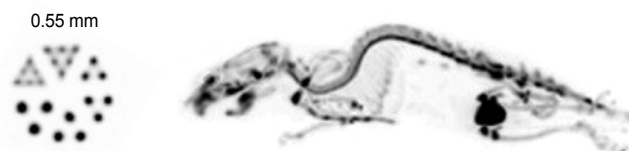
Because animal PET as we know it has reached its limits

Animal PET imaging, as we know it based on coincidence detection has reached its limits. After twenty years of scaling down clinical human PET scanners, it is no longer possible to adopt the improvements of clinical PET to small animal imaging. All technological advances in human PET stem from the implementation of Time-of-Flight PET (TOF-PET). Unfortunately, the huge size difference between men and mice creates unsurmountable challenges for the implementation of TOF for small animal PET imaging. All crystal-based detectors are too slow to determine with any precision the location of a positron annihilation event within a small animal. Therefore, and since image magnification cannot be used with coincidence-based collimated PET, the resolution depends on the best resolution that can be achieved within the crystal detector itself, approx. 1.0 to 1.2 mm. As illustrated on the previous page, many attempts to improve on this using coincidence detection have failed.

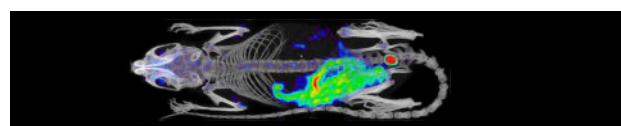
U-PET image examples including dynamic low-dose whole-body PET



Whole-body low-dose dynamic PET imaging of ^{18}F -RGD [13] on the left, and ^{89}Zr -RGD peptide imaging on the right (image courtesy of University of Saskatchewan).



^{18}F -NaF images of a phantom and a mouse (image courtesy NIRS-Japan).



^{18}F -FDG PET/CT dynamic evolution of inflammatory foci for imaging dengue virus infection and treatment response [14].

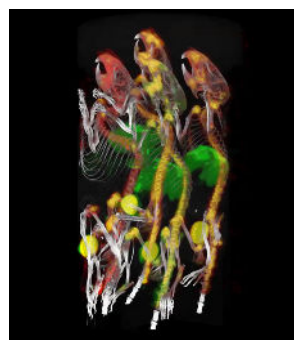
Because we need U-PET to reach the visual acuity of human PET

To reach the visual acuity required for translating data from mouse images to those of human PET scanners, MILabs took a page out of the playbook of the microscopy evolution. Rather than trying to image a wide field without magnification and with a lot of unwanted noise contribution from surrounding and scattered activity, MILabs took the “confocal style” approach to develop U-PET. By quickly moving a focused magnified field-of-view through an animal, images can be obtained with a resolving power unattainable in conventional preclinical PET systems. The result: volumetric image resolutions can be ten times improved and interfering noise reduced tenfold.

The advantages of U-PET

The following table illustrates in more detail how U-PET eliminates most of the current shortcomings of conventional coincidence-based PET systems:

System capabilities	MILabs U-PET	Conventional PET
Elimination of noise and bias from randoms	Yes	No
Low scatter noise and bias	Yes	No
Demagnification of detector blurring	Yes	No
Photon acollinearity elimination	Yes	No
Uniform sub-mm resolution	Yes	No
Positron range elimination: sub mm resolution ^{124}I , ^{82}Rb imaging	Yes	No
α and β radiotherapy imaging	Yes	No
Simultaneous PET/PET ($^{18}\text{F}/^{124}\text{I}$, $^{82}\text{Rb}/\text{FDG}$)	Yes	No
Simultaneous sub-mm PET/SPECT	Yes	No



“The resolution and sensitivity of MILabs U-PET are exquisite. For instance, we are capable of running dynamic PET at high temporal and spatial resolution with low < 5 MBq activity levels, while tumor heterogeneity can still be observed with only 0.01 MBq delivered to the tumor” (The impact of U-PET on their research, Lead Biological Scientist at University of Oxford).

Simultaneous multi-isotope PET and SPECT imaging



VECToR⁶CT: The system with unique tri-modality omni-imaging capabilities

Unmatched PET/SPECT/CT performance: sub-mm PET and SPECT resolution even when imaged simultaneously in space and time. Anatomical co-registration with fast, ultra-high resolution, low-dose X-ray CT.

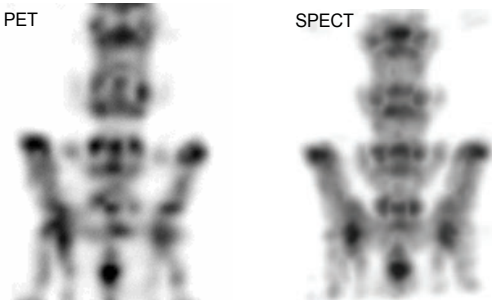
Exclusive spatial and temporal co-registration of PET and SPECT images. Sub-mm imaging of co-injected PET and SPECT tracers enables exploring different physiological and molecular functions at the same time, under identical physiological and physical conditions. Allows switching between SPECT- and PET-based approaches at any time.

Unmatched molecular sensitivity for PET and SPECT imaging enables targeting specific biological processes with low-dose PET and/or SPECT tracers.

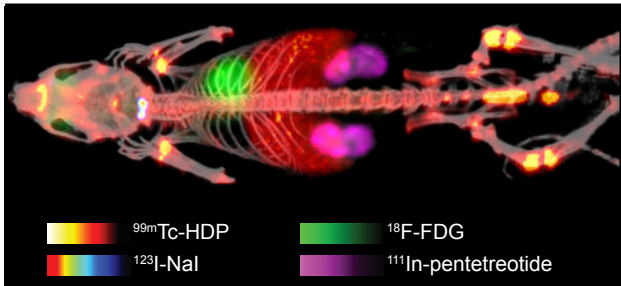
Dynamic PET and SPECT: excellent whole-body (< 8 s) and organ (< 1 s) time resolution.

Advanced theranostics applications: the ability to image all high-energy isotopes at sub-mm resolution including radiotherapeutic Auger e⁻, α-emitters, and β-emitters such as ¹³¹I [15], ¹⁸⁸Re [20], ²¹³Bi [16], ²⁰⁹At [21], ²²⁵Ac [17], and ²²³Ra [22] can be combined with simultaneous imaging of their therapeutic effect using almost any other PET or SPECT tracer.

Simultaneous PET/SPECT of co-injected tracers

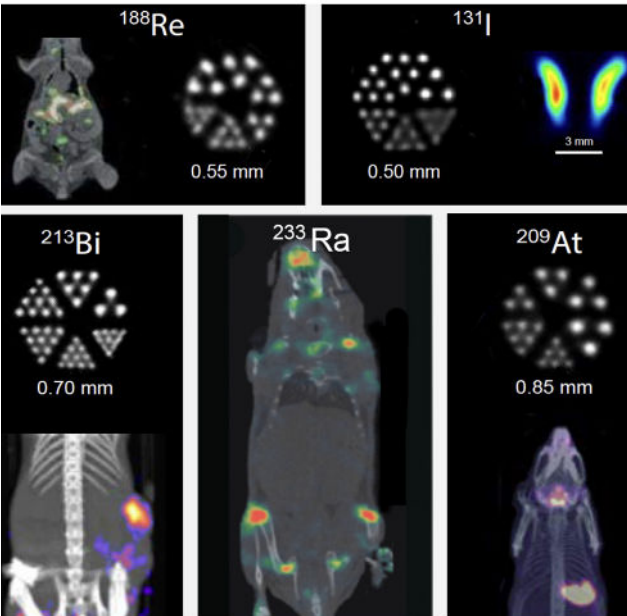


VECToR: ¹⁸F-NaF PET and ^{99m}Tc-HDP SPECT co-registered in time and space obtained in a single mouse scan.



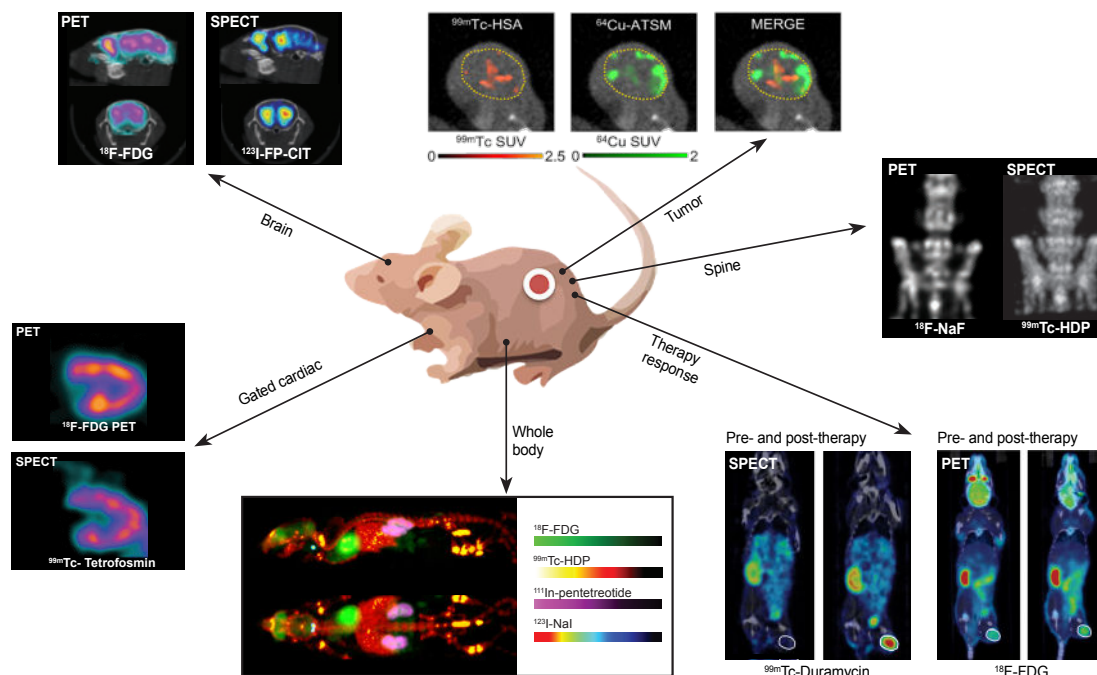
Simultaneous SPECT and PET imaging of four tracers [23].

Radiotherapy and theranostics



With VECToR⁶CT, you can investigate virtually any radiotherapy or theranostic isotope, even those with high energies, at sub-mm resolution

Co-injected tracers with spatial and temporal co-registration



Examples of nuclear imaging beyond the capabilities of any other system:

- Theranostics: high-energy SPECT with PET/SPECT treatment follow-up [15, 16].
- Comparing immuno-PET (e.g. ^{89}Zr) and immuno-SPECT (e.g. ^{111}In).
- DAT-SPECT and FDG-PET to differentiate typical from atypical parkinsonian syndromes.
- Combined PET/SPECT monitoring of pharmacokinetics and treatment response (e.g. using $^{99\text{m}}\text{Tc}$ -Duramycin or ^{18}F -FDG) [24].
- Simultaneous ^{124}I -immuno-PET and tumor ^{18}F -FDG/FLT scans.
- Simultaneous $^{89}\text{Zr}/^{111}\text{In}$ tumor heterogeneity imaging: “*in vivo* immunohistochemistry” [25].
- Simultaneous perfusion and metabolism imaging to enhance the diagnostic value of cardiac imaging.
- Alzheimer imaging: SPECT (e.g. $^{99\text{m}}\text{Tc}$ -HMPAO) with PET (e.g. ^{11}C -PIB).
- Simultaneous vascularity ($^{99\text{m}}\text{Tc}$ -HSA) and hypoxia (^{64}Cu -ATSM) imaging in tumors [26].
- Single scan comparisons of “identical” SPECT and PET tracers: e.g. labeled with ^{67}Ga vs ^{68}Ga , $^{125}\text{I}/^{123}\text{I}/^{131}\text{I}$ vs ^{124}I , ^{111}In vs ^{64}Cu .
- Comparing long-lived ^{124}I to short-lived ^{18}F tracers using both spatial and temporal coregistration.
- ^{82}Rb MPI PET versus dynamic SPECT with e.g. $^{99\text{m}}\text{Tc}$ -sestamibi dynamic SPECT.

The unique application advantages of Ultra-high-definition PET

- Superior Resolution: ~10-fold volumetric resolution improvement for ^{18}F and > 50-fold for ^{124}I and ^{82}Rb .
- Linear NECR response: from < 1 MBq to 50 MBq; noise overwhelms specific signals only at high activity levels.
- Uniform image quality over the selected FOV, inclusive for whole-body imaging.
- Simultaneous dual-isotope PET for selected isotope pairs such as $^{124}\text{I}/^{18}\text{F}$ and $^{82}\text{Rb}/^{18}\text{F}$
- Theranostic imaging for alpha and beta emission radiotherapies with energies up to 1,000 keV
- Simultaneous multi-isotope PET/SPECT imaging (with SPECT option)

20 ^{188}Re image performance assessment using small animal multi-pinhole SPECT/PET/CT system, P. L. Esquinas et al., Physica Medica, 2017
 21 Evaluation of ^{209}At as a theranostic isotope for ^{211}At -radiopharmaceutical development using high-energy SPECT, JR Crawford et al., Phys. Med. Biol., 2018
 22 First Whole-Body Three-Dimensional Tomographic Imaging of Alpha Particle Emitting Radium-223, D. S. Abou et al, bioRxiv Sep. 26, 2018
 23 Integration of Small Animal SPECT and PET with Other Imaging Modalities, P.E.B. Vaissier et al., TvNG 2013
 24 $^{99\text{m}}\text{Tc}$ -Duramycin SPECT imaging of early tumor response to targeted therapy: a comparison with ^{18}F -FDG PET, F. Elvas et al., Nucl. Med., 2016
 25 Dual-isotope imaging allows *in vivo* immunohistochemistry using radiolabelled antibodies in tumors, JC Knight et al., Nucl. Med. Biol. 2019
 26 *In Vivo* Simultaneous Imaging of Vascular Pool and Hypoxia with a HT-29 Tumor Model: the Application of Dual-Isotope SPECT/PET/CT, N. Adachi et al., IJSBAR 2016

Exclusive dual-use: Duet Optical Imaging module



Duet Optical Imaging module

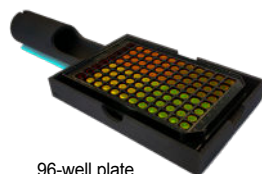


More Throughput. More Capabilities. More Flexibility.

- Use your Optical Imaging System as never before. On a bench, with a docking station for high throughput 2D imaging of up to ten mice at a time. Alternatively, on an automated X-ray CT platform for automated 2D or deep-tissue, quantitative 3D tomographic optical/CT imaging - at any time upgradeable with translational PET & SPECT.
- With an industry-leading optical camera, the Duet Optical Imaging module is designed to offer the ultimate in high-sensitivity, low noise performance, over an extended wavelength range of 400-900 nm.
- The large area sensor features multi-stage thermoelectric cooling, enabling the camera to operate at an unrivaled -100°C for single photon sensitivity at $> 90\%$ quantum efficiency. However, more importantly, it comes with an optimal balance between resolution and dynamic range with 1024×1024 pixels to ensure fast readout at very low noise - all requirements for new applications such as deep-tissue 3D imaging, which increases sensitivity by one to two orders of magnitude (2048×2048 pixels available as an option but not recommended).
- Suited for optical reporters across the blue to near infrared wavelength region: the system uses a computer-controlled filter wheel with narrow band-pass emission filters for multi-spectral bioluminescence, Cherenkov and fluorescence imaging.
- For fluorescence mode emission, a broadband halogen light source uses a set of computer-controlled narrow band-pass excitation filters.

Unmatched OI versatility

U-OI can scan up to five mice in a single system pass in the standalone system, three in the integrated system, and can also scan 96-well plates and microscopy slides. It is upgradeable to U-OI/CT and 3D fluorescence and bioluminescence tomography (FLT/CT and BLT/CT).

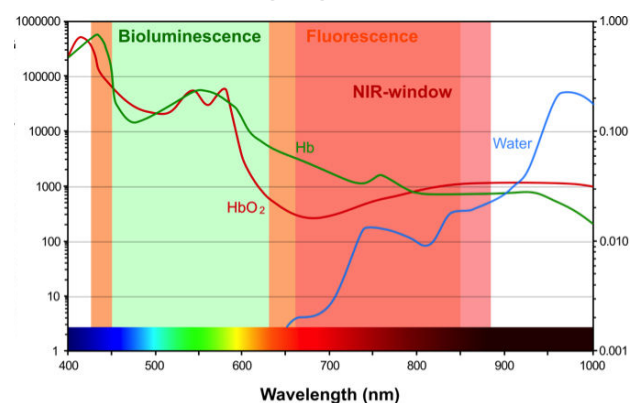


96-well plate



5-mouse holder

Bioluminescence, Fluorescence and Cherenkov imaging



Since the Duet Optical module's wavelength range covers Cherenkov Luminescence (CL) or Secondary Cherenkov-induced Fluorescence Imaging (SCIFI), the system can be used to image radionuclides that emit charged particles such as β^- . This ability further complements the theranostic applications of VECTOr⁵ by enabling e.g. ^{90}Y radiotherapy imaging.

A scalable & highly flexible optical imaging platform

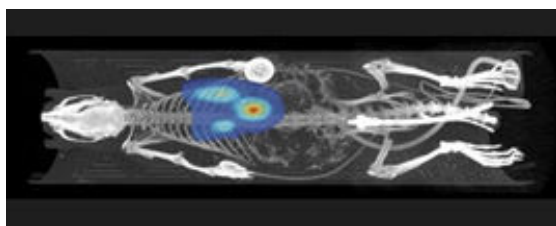
Imalytics Spectral Unmixing

By acquiring images with multiple narrow-band excitation and emission filters, the acquired spectra of multiple probes can be unmixed using the spectral unmixing feature of MILabs' optional Imalytics Software [27].

Enhancing OI with Diagnostic CT

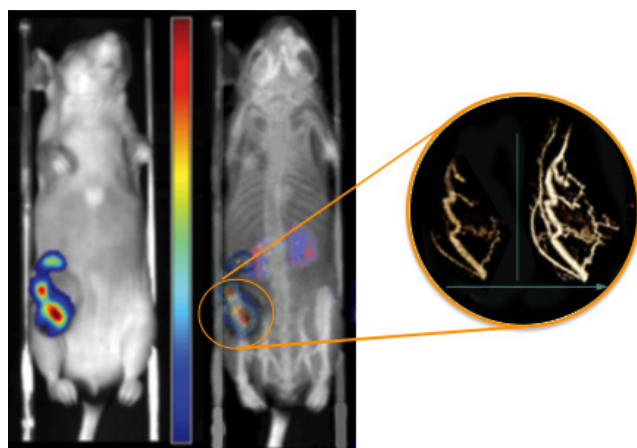
As the only optical imaging system with an integrated CT with diagnostic capabilities, VECTor® enables to:

- Use Perspective MIP imaging to fuse 2D bioluminescence or fluorescence images with 3D X-ray CT anatomical images (e.g., bioluminescent signals from liver tumors).



Anatomical co-registration of 2D optical and 3D CT using Perspective Maximum Intensity Projections (MIP)

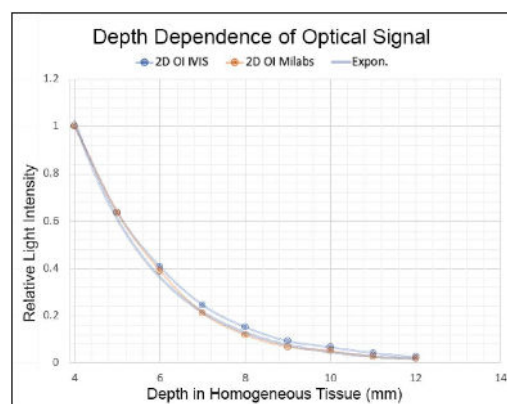
- Use DCE-CT to determine whether light intensity changes in tumors are actually due to treatment effects or caused by acute hypoxia due to rapid growth or inadequate blood supply.



Beyond Optical/CT co-registration: the in-line Diagnostic CT enables to look at tumor vascularity to validate optical signal response to treatment.

Sensitivity of 2D Optical Imaging

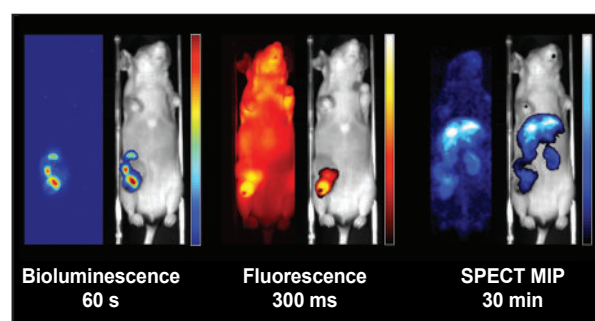
Because of the depth dependence of 2D optical signals for in-vivo imaging, it is difficult and not always relevant to compare the optical sensitivities of different systems using light intensities measured at the surface of animals. The following graph illustrates that the sensitivity of MILabs' 2D OI system at different tissue depths correlates well with the established IVIS system. However, it also abundantly illustrates that no matter what system used, 3D tomography (either optical or nuclear) is required for true quantitation.



Exponential decline of measured 2D OI signal in homogeneous tissue phantom. Relative intensities normalized to one at 4 mm depth [28]

Enhancing OI with Nuclear Images

In a single scanning session, and with a single dose of anesthesia, VECTor®OI/CT enables complementary registration of bioluminescent, fluorescent, CT, PET, and SPECT images.



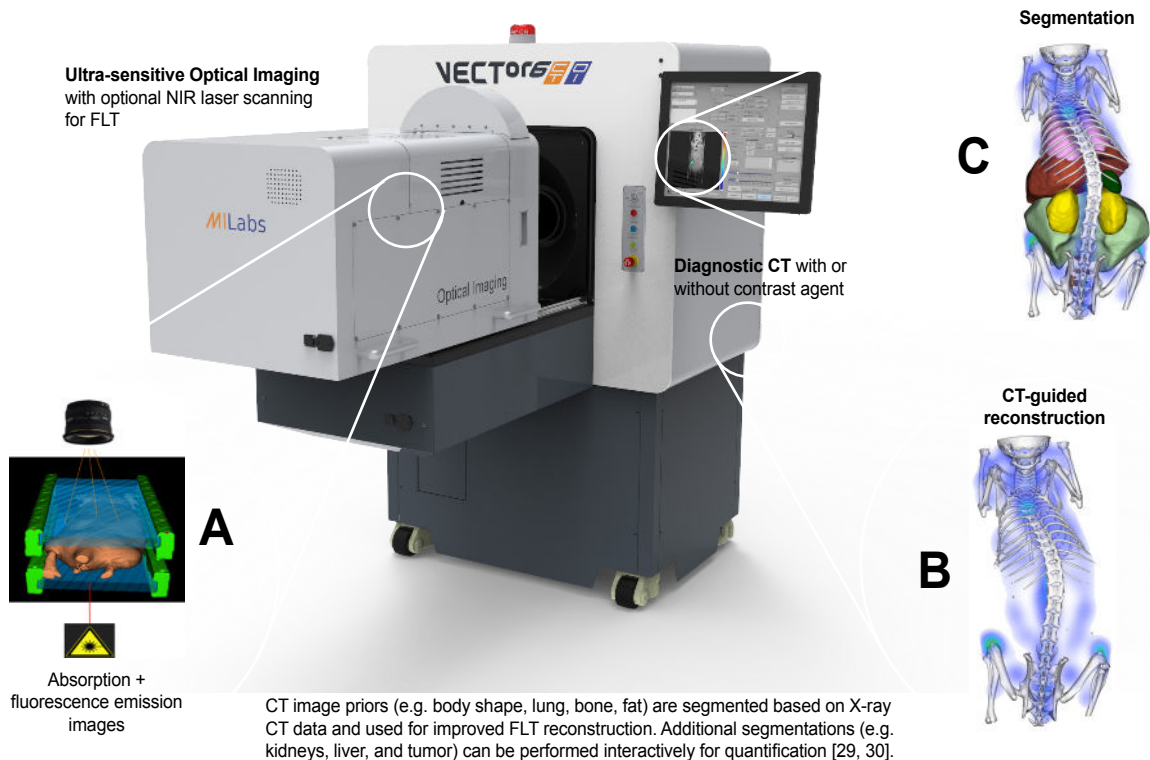
Example of complementary registered bioluminescent, fluorescent, and nuclear images in one scanning session with a single dose of anesthesia [28].

27 Data Curation for Preclinical and Clinical Multimodal Imaging Studies. Grace Gyamfuah Yamoah et al. Molecular Imaging and Biology, March 2019

28 U-SPECT-BioFluo: An Integrated Radionuclide, Bioluminescence, and Fluorescence Imaging Platform. M.N. van Oosterom, et al, EJNMMI Research, 2014

A breakthrough in hybrid Optical/CT tomography

The powerful combination of two industry-leading modalities with the best image analysis and segmentation software (Imalytics Preclinical [28]) enables quantitative optical tomography imaging with fluorescence and bioluminescence.

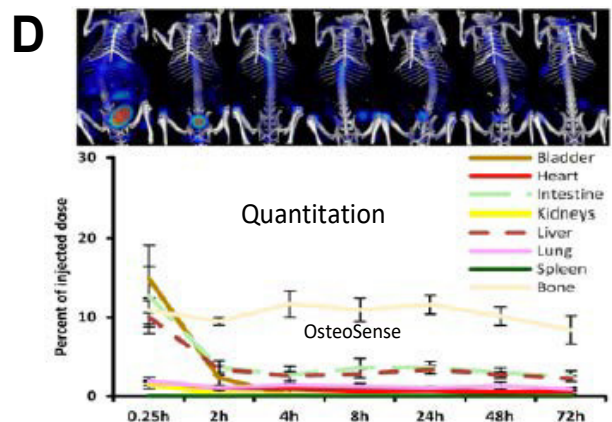


Hybrid FLT/CT

Hybrid FLT/CT goes beyond the capabilities of stand-alone FLT (also called Fluorescence Molecular Tomography, FMT) by using the heterogeneous optical absorption and scattering maps of the animal for improved fluorescence reconstruction [31].

Hybrid BLT/CT

Hybrid BLT/CT uses the same 3D bioluminescent light correction approach as FLT by using segmented CT priors, but reconstructs the initial 3D BLT image by imaging bioluminescence at multiple wavelengths [32].



29 Imalytics Preclinical: Interactive Analysis of Biomedical Volume Data, F. Gremse et al., Theranostics 2016

30 Method for reconstructing the image of an object scanned with a laser imaging apparatus, Rohler et al., US patent 6,130,958; 2000

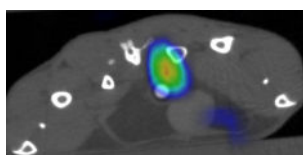
31 Spectrally resolved bioluminescence optical tomography, H. Dehghani et al., Optics Letters, 2006

32 Non-Invasive Optical Imaging of Nanomedicine Biodistribution, S. Kunjachan, F. Gremse et al., ACS Nano. 2013

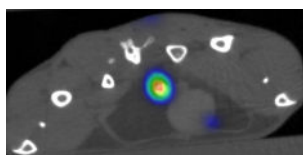
Quantitative Optical Tomography that rivals PET/SPECT

Why hybrid FLT/CT is superior?

Light scattering and absorption are heterogeneous in the body of an animal. These light characteristics depend on the “darkness” of the organ/tissue from where the optical probe emits light. As illustrated below, accurate optical tomography is not possible without correcting for these heterogeneities.



Simple 3D OI/CT (FMT/CT)



Hybrid FLT/CT

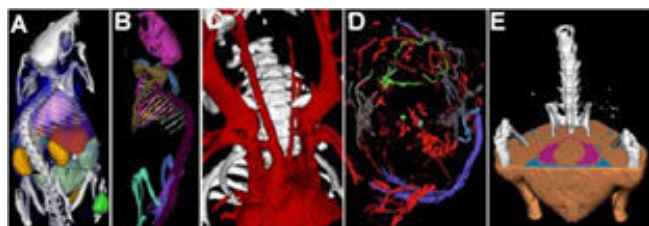
Top: Erroneous reconstruction results of simple 3D OI/CT (or FMT/CT) imaging. This method does not account for the heterogeneous light absorption within different tissues.

Bottom: MI Labs' hybrid FLT/CT system using prior CT information to reconstruct the fluorescence images based on heterogeneous absorption. This results in a better localization and intensity, which is further explained by Gremse et al. [33].

Imalytics Preclinical segmentation and analysis software

Imalytics Preclinical [28] image segmentation software is excellent for preclinical applications. Results have been published in numerous peer-reviewed articles for:

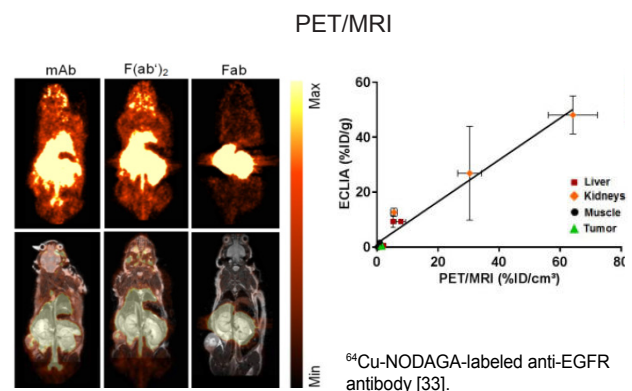
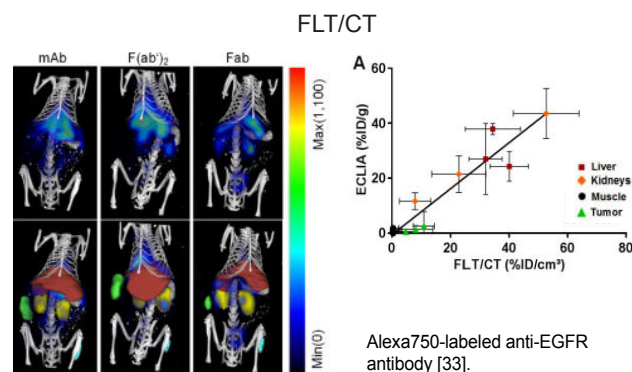
- Organ segmentation for biodistribution studies
- Bone segmentation and labeling
- Segmentation of blood vessels
- Fat segmentation (visceral and sub-cutaneous)
- Hybrid FLT/CT & BLT/CT analysis and renderings



A. Organ segmentation, B. Bone segmentation, C. Blood vessels and bone segmentation, D. Vascularity segmentation E. Fat segmentation.

Hybrid FLT/CT versus PET/MRI

Hybrid FLT/CT and PET/MRI images and analyses show the biodistribution in separate cohorts of nude mice bearing subcutaneous tumors, two hours post injection of anti-EGFR compounds. *In vivo* analyses correlated well with *ex vivo* quantitation for both modalities [34].



MI Labs U-OI features	OI	Hybrid OI/CT
2D bioluminescence, fluorescence, and Cherenkov imaging	✓	✓
Maximum number of animals	10	3
96-well plate imaging	✓	✓
Spectral unmixing	Optional*	Optional*
In-line X-ray CT	Optional	✓
3D FLT/CT and BLT/CT	N/A	Optional*
Imalytics Preclinical	Optional	Optional

* Requires Imalytics Preclinical software

33 Absorption reconstruction improves biodistribution of fluorescent nanoprobes using hybrid fluorescence tomography, F. Gremse et al., Theranostics 2014.

34 Comparison of the accuracy of FMT/CT and PET/MRI for the assessment of antibody biodistribution in squamous cell carcinoma xenografts, C. Hage et al., J. Nucl. Med., 2017

The only CT system that will fit your needs - Today, tomorrow and beyond...



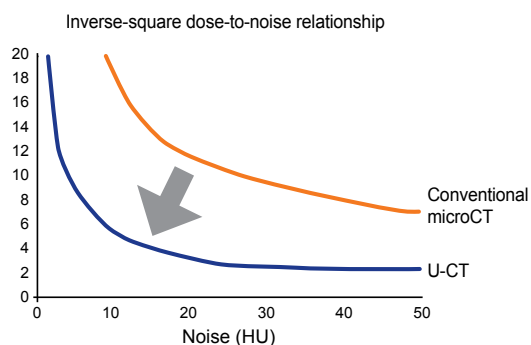
Industry-leading 3D/4D adaptive X-ray CT

MILabs' U-CT is extremely fast and offers ultra-high resolution at low X-ray doses enabling demanding multispecies 3D/4D *in vivo* applications, including dynamic contrast-enhanced (DCE) CT imaging.

Many steps ahead in preclinical CT:

- Stand-alone or integrated CT, field-upgradable with SPECT, PET, and 2D/3D Optical imaging
- With variable isotropic voxel resolution down to 2.4 μm , the U-CT system is also suited for *ex vivo* applications
- Ultra-fast whole body scanning down to 5 s
- Ultra-low dose, down to < 2 mGy whole-body mouse

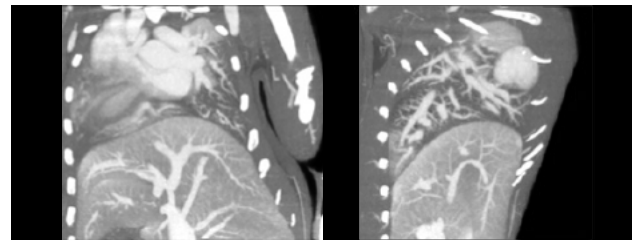
Unrivalled *in vivo* dose efficiency



By combining cutting-edge CMOS technology with a thick scintillation layer and on-the-fly readout, U-CT sets new low-dose benchmarks without compromising image quality.

Cardiac- and respiratory-gated CT

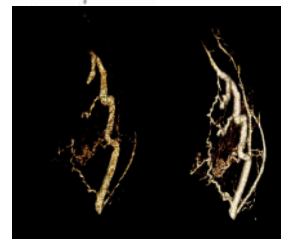
Accurate CT reconstruction in cardiac and respiratory research requires motion elimination from the heart and diaphragm. U-CT uses dual-gated acquisitions and freeze-frame to avoid image blurring.



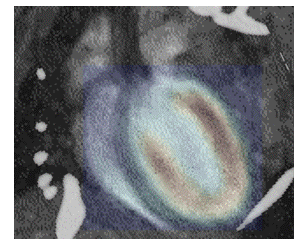
Respiratory- and cardiac-gated "freeze-frame" image.

Unique DCE-CT & PET/SPECT/CT

With fast, high-resolution, and optional dual-energy acquisitions, U-CT is uniquely suited for diagnostic DCE-CT applications. Moreover, MILabs' unique attenuation correction enables hybrid PET/SPECT/CT imaging, even for gated acquisitions [35].



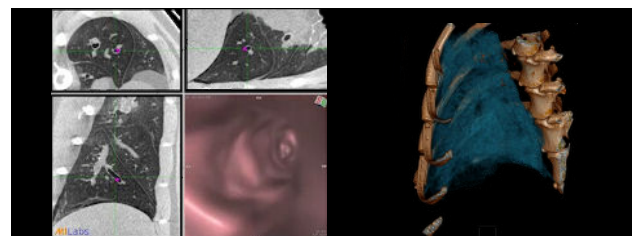
DCE-CT* of tumor vasculature [36].



U-PET/CT ^{18}F -FDG and Exiton

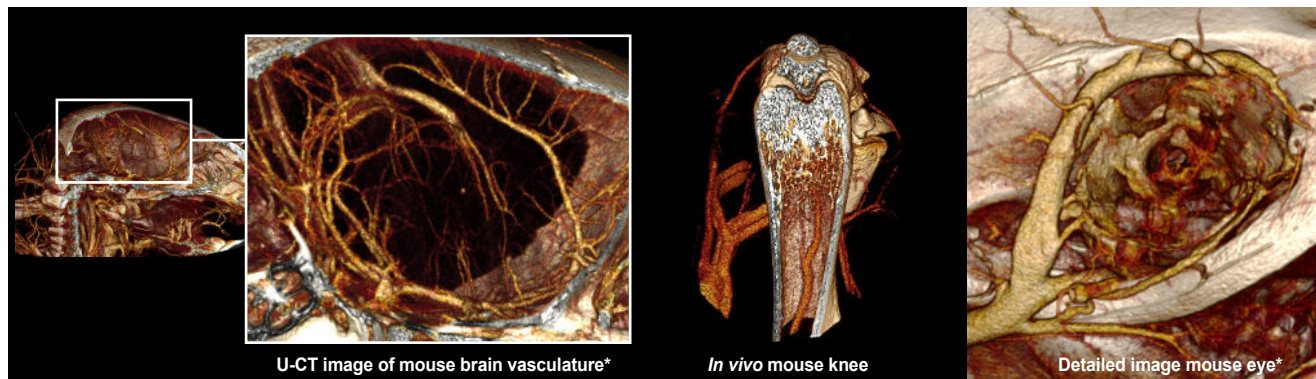
From small specimen to larger animals

MILab's U-CT is a true multispecies microCT system capable of imaging *ex vivo* specimen as well as a range of *in vivo* animals. The FOV is large enough to image entire mice, rats and optionally, ferrets and rabbits.



Virtual lung endoscopy of a ferret.

MILabs' U-CT system will adapt to your applications so you don't have to adapt your research. Why invest in multiple systems for speed and resolution, for *in vivo* and *ex vivo* applications, or for molecular CT with SPECT, PET or OI, if you can start your research with a single U-CT today and expand by adding modalities over time?



Autonomous X-ray CT gantry,
the key reason for maintaining
true diagnostic CT performance
on a scalable multimodal platform

Molecular CT
w/SPECT, PET, OI

Mice, rats &
rabbits plus
EXIRAD-3D
specimen

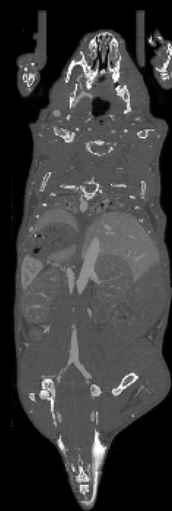
Ultra-High Resolution
2.4 micron voxel size

Fast scanning
whole-body in 5 s

Low dose
< 2 mGy whole-body



Dual-gated cardiac CT*



Whole-body mouse*

Other advantageous options are:

- Multi-mouse CT for up to 4 mice simultaneously
- Sensor-free image based cardiac and respiratory gating, required for simultaneous multi-animal imaging
- Ultra-fast fluoroscopic imaging with up to 66 frames per second (no separate fluoroscopy device needed)
- Virtual endoscopy for mice up to rabbits, including non-invasive bronchoscopy

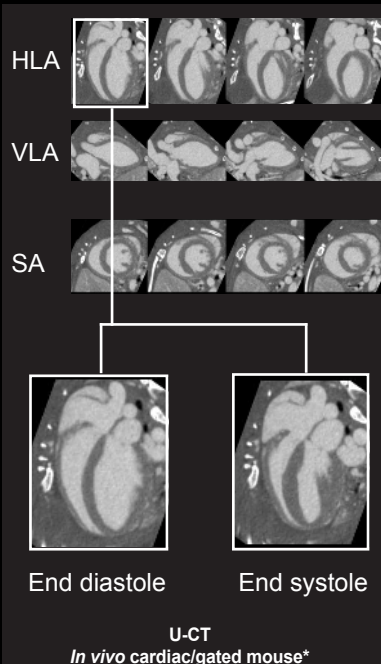
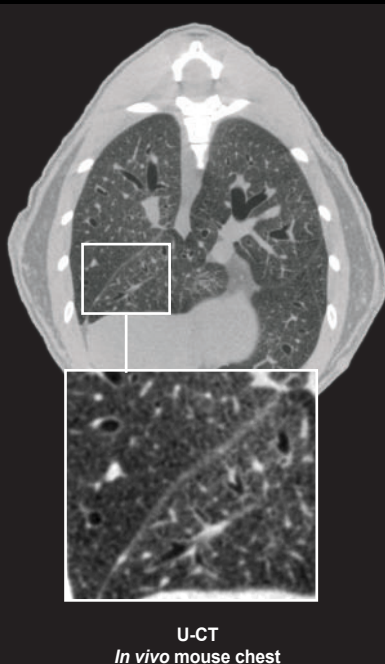
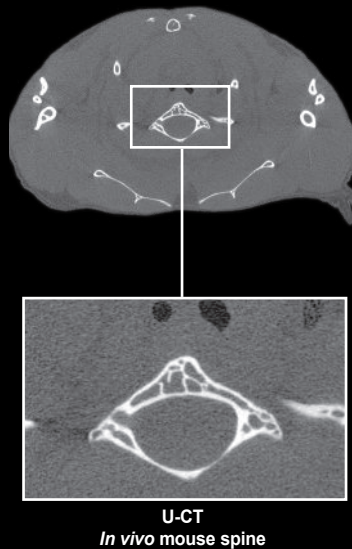
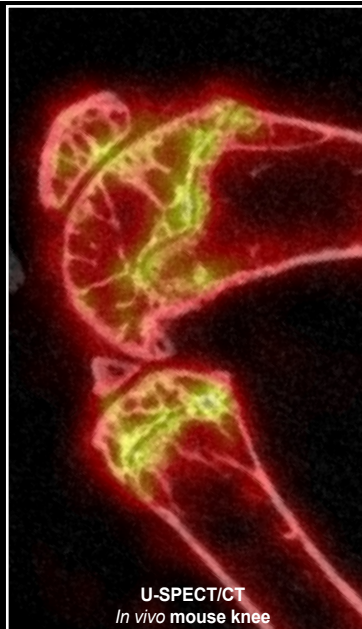
35 Oral contrast enhances the SPECT/CT resolution of in-life NIS reporter gene imaging, L. Suksanpaisan et al, Cancer Gene Therapy, 2013

36 Courtesy of Dr. Sean Smart and Dr. Veerle Kersemans, CRUK/MRC Oxford Institute for Radiation Oncology, University of Oxford

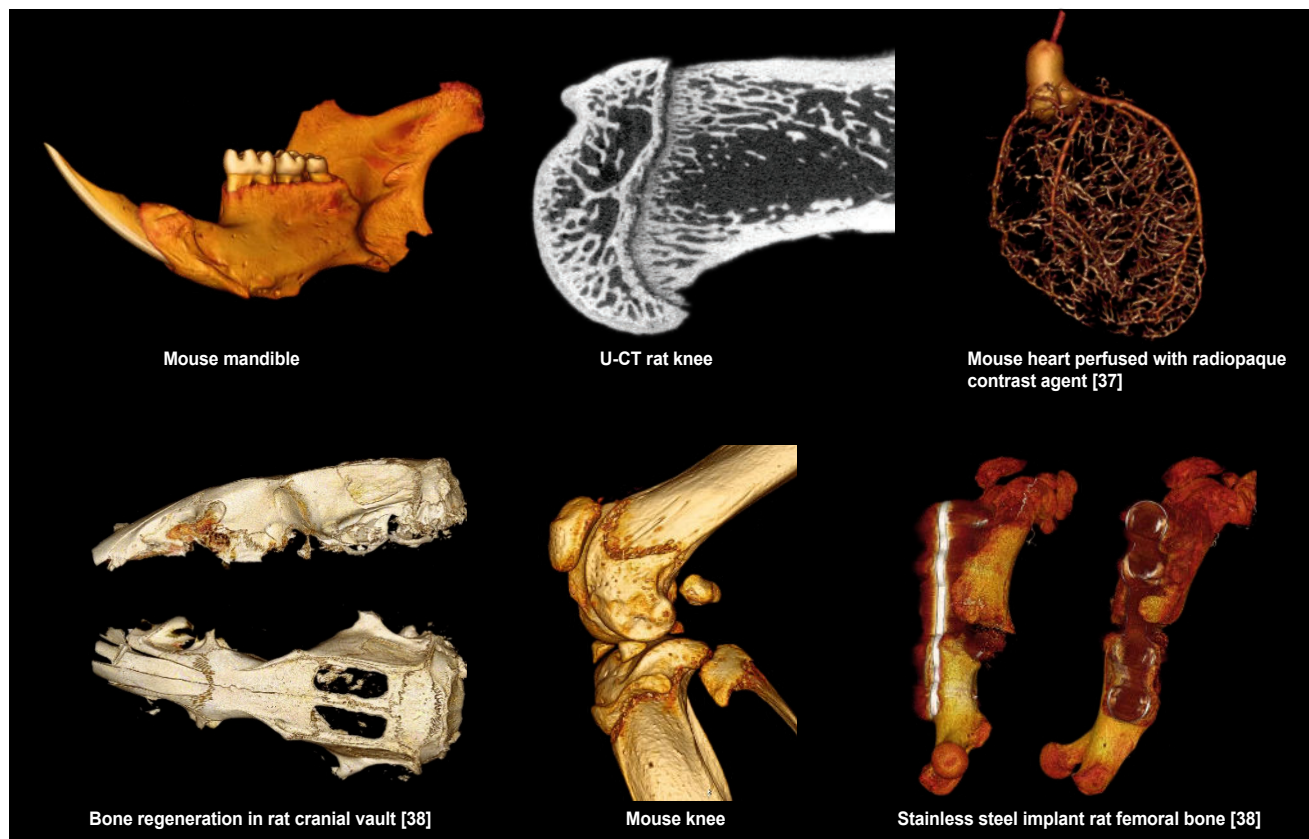
*Imaged using ExiTron nano 12000 – nanoPET Pharma GmbH.

Best resolution, highest speed, and lowest dose for *in vivo* imaging, plus scaleable to image small specimen as well as larger animals including rabbits. In addition, upgradable with in-line OI and/or PET and SPECT plus the unique ability to handle DCE-CT and attenuation-corrected PET/CT and SPECT/CT, including gated multi-modality acquisitions. The system allows for dual-gated cardiac-

and respiratory imaging with freeze-frame acquisitions and is equipped with an on-screen dose estimator. Add to this the flexibility to tailor configurations to your research needs and you will agree that there are no other X-ray CT systems that can bring as much diagnostic CT power to your preclinical imaging applications.



MILabs' images are fully compatible with existing software based on international image format standards and work seamlessly with post-processing image analysis software packages such as PMOD, Imalytics Preclinical, VivoQuant, Amira, ImageJ, and OsiriX. *In vivo* images acquired by U-CT can be analyzed and quantified in high detail using the user's favored software. For example, bone parameters like trabecular thickness and structure model index can be easily extracted. Additionally, images can be exported to STL file format for detailed surface rendering and analysis while all packages allow advanced image visualization in 2D, 3D, and even 4D.



X-Ray CT system/module	Standard	HR	UHR	XUHR
Smallest isotropic voxel size	30 μm	10 μm	4 μm	2.4 μm
Focal spot size	25 μm	25 μm	7 μm	7 μm
Total body mouse scanning time	5 s	5 s	5 s	5 s
Total body mouse dose	< 5 mGy	< 5 mGy	< 2 mGy	< 2 mGy
Maximum field of view (FOV)	$\varnothing$ 75 mm	$\varnothing$ 75 mm	$\varnothing$ 75 mm	$\varnothing$ 75 mm
Focused scanning		✓	✓	✓
Focused scanning in <i>ex vivo</i> specimen holder				✓
Circular and helical scanning	✓	✓	✓	✓
Mouse	✓	✓	✓	✓
Rat	✓	✓	✓	✓
Medium sized animal (FOV $\varnothing$ 130 mm)	Optional	Optional	Optional	Optional
Dual-energy CT	Optional	Optional	Optional	Optional
Dynamic (multi-frame) imaging	Optional	Optional	Optional	Optional
Respiratory and cardiac gating	Optional	Optional	Optional	Optional

37 Sample courtesy of Prof. A.J. Sinusas, MD and Zhenwu Zhuang, MD, MS, Yale School of Medicine, USA

38 Samples courtesy of Prof. R. Jayakumar, Amrita Center for Nanosciences and Molecular Medicine, India

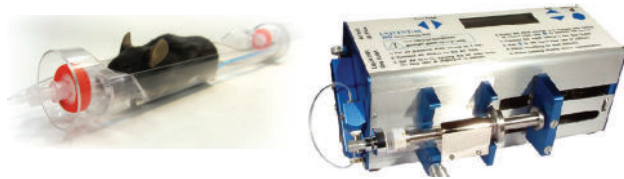
Dedicated animal cells and monitoring systems

Animal beds and cells

MILabs' transparent animal imaging beds can be easily converted to closed imaging U-cells for pathogen-free imaging. They are equipped with anesthesia cones and heater pads for optimal animal comfort. The beds can easily be mounted onto, and removed from, MILabs' systems and also facilitate smooth transfer to other modalities such as MRI. The beds allow for open access to the animal during acquisition or can be closed. There is continuous visual monitoring of the animal by four optical cameras. Custom beds are available upon request.

Anesthesia

Animals can be anesthetized with an optional gas anesthesia system with built-in vaporizer and flow meter. Precise automatic control of anesthetic and air minimizes anesthetic consumption.

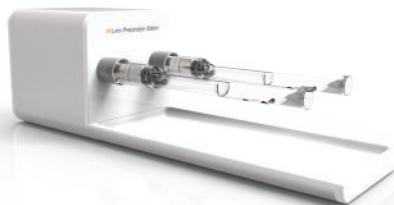


Gating, heating and monitoring

An integrated system is available for animal monitoring, automatic animal temperature regulation, and multiple physiological event triggering. This enables flexible triggering, e.g. combined respiratory and cardiac gating.



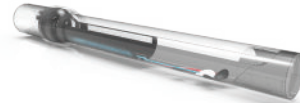
Docking station



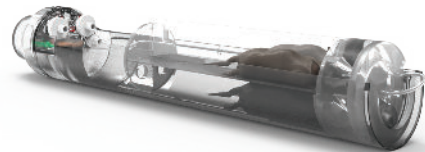
Animals can easily be positioned, injected and prepared using movable docking stations. These can be used either on the bench or inside a fume-hood.



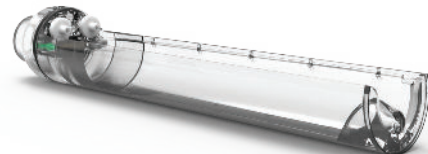
Mouse bed optimized for comfort, reproducibility, and reliability.



U-cell providing closed environment, suited for immuno-compromised or contagious mice.



Multi-mouse bed, for scanning 2, 3 or 4 animals simultaneously.



Rat bed with anesthesia nose cone, heating and gating connections.



Ferret bed with anesthesia nose cone, heating and gating connections.



Medium size animal bed for animals up to 5 kg.



Holder for stand-alone OI and hybrid OI/CT (max. three mice).



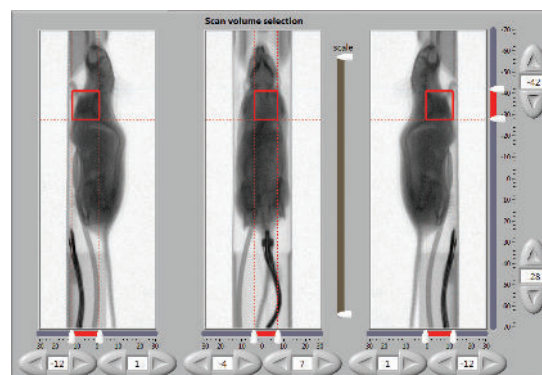
Animal holders for hybrid FLT/CT and BLT/CT imaging.

Image acquisition, reconstruction, and visualization

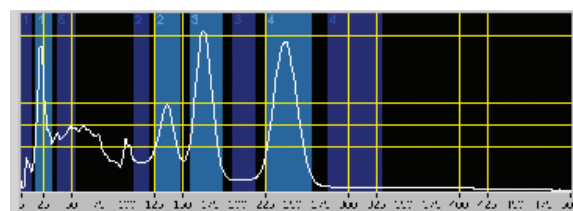
MILabs' systems are provided with advanced proprietary acquisition, reconstruction, fusion, and analysis software. The software is intuitive and extremely user-friendly, so imaging with MILabs's systems can typically be mastered by a technician or a student in a single day.

GUI-based list-mode acquisitions

The patented acquisition graphical user interface (GUI) is displayed on a touch screen. Unique interactive sliders enable easy and accurate selection of the field-of-view in three orthogonal directions. The scan volume is selected from either three X-ray or optical camera views and helps improve signal strength from the tissues of interest [39]. True List Mode (TLM) acquisitions facilitate quantitative multi-isotope imaging with a virtually unlimited number of energy windows. With the VECTor[®]CT, a PET isotope can easily be imaged simultaneously with a combination of several different SPECT isotopes. Furthermore, with TLM, background windows can be defined retrospectively to accurately correct for scatter and spill-over effects, and stored accumulated counts can be fractionated to study the effect on image quality, thus enabling dose planning without rescanning.



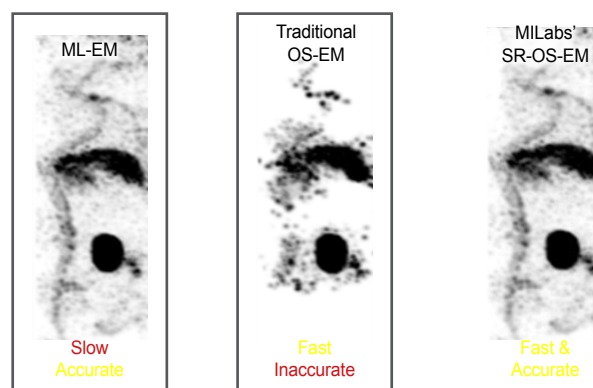
Field-of-view selection from total body and organs using optical and/or X-ray CT images exploits MILabs' fast-focused 3D acquisition [37].



Exclusive True List Mode acquisition facilitates quantitative multi-isotope imaging and dose-planning.

Reconstruction software

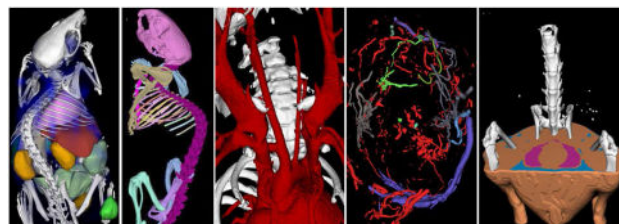
Novel reconstruction algorithms such as model based ML-EM, OS-EM, Pixel-based OS-EM (P-OS-EM) [40] or statistically regulated OS-EM (SR-OS-EM) [41] are used for SPECT and PET reconstruction. These reconstruction algorithms are up to 30 times faster than traditional OS-EM while accuracy is improved [41].



MILabs' SR-OS-EM [39] provides fast high-fidelity image reconstruction

Visualization, analysis and storage

An enterprise-wide license for the visualization and analysis software platform comes with every system installation. Image data storage and handling is in full accordance with the latest DICOM standard. Other data formats include BMP, TIFF, JPG, AVI, and NIFTI which makes import into image analysis software packages such as Imalytics Preclinical, PMOD, InviCRO-VivoQuant, Amira, ImageJ, OsiriX and MILabs' BoneJ analysis software easy.



Imalytics Preclinical [27] software for hybrid FLT/CT and BLT/CT. From left to right: biodistribution, bone segmentation, vessel segmentation, tumor vascularity, and fat segmentation.

39 Targeted Multi-pinhole SPECT. W. Branderhorst, B., et al., Eur. J. Nucl. Med. Mol. Imaging, 2011

40 Pixel-based Subsets for Rapid Multi-pinhole SPECT Reconstruction. W. Branderhorst, et al., Phys. Med. Biol., 2010

41 Similarity-regulation of OS-EM for accelerated SPECT reconstruction P. Vaissier, et al., Phys. Med. Biol. 2016

MILabs' ground breaking imaging platforms have received several international industrial innovation and scientific awards, e.g. from Frost & Sullivan, the Society of Nuclear Medicine and the World Molecular Imaging Society. Each year MILabs significantly invests in new technologies such as software and hardware upgrades to ensure that MILabs' customers can continuously expand their cutting edge research.



Frost & Sullivan: Medical Innovation Product Award



Dutch Science Foundation: Physics Valorization Prize 2013



Society of Nuclear Medicine: Animal Image of the Year



World Molecular Imaging Society



Soc. Nucl. Med. & Mol. Im.: Hoffman Memorial Award 2017



World Molecular Imaging Society

MILabs is committed to supporting the scientific and medical community by providing the best possible imaging solutions on the market. MILabs is proud that world leading academic partners have chosen MILabs' systems. MILabs collaborates with many institutes worldwide through joint grant applications, sharing of expertise, organizing workshops and developing customized imaging solutions with a single goal in mind: to facilitate and perform the best research and diagnostics.

